



Option Parameter	Relay Output	ALARM Function Connection Detail for NXTSD104 Only		
17.1	1	Display, Low or Line voltage	PR1	Common
			PR2	Normally Open
			PR3	Normally Closed
17.2	2	Display, Low or Line voltage	PR4	Common
			PR5	Normally Open
			PR6	Normally Closed
17.3	3	Display, Low or Line voltage	PR7	Common
			PR8	Normally Open
			PR9	Normally Closed
17.4	4	PPC6000, PE1 PE4 Line voltage output ONLY		
17.5	5	N/A		
17.6	6	N/A		
17.7	7	NXDBVSD, PZ15, PZ16, Low voltage <50V AC/DC, 200mA		
17.8	8	NXDBVSD, PZ17, PZ18, Low voltage <50V AC/DC, 200mA		
17.9	9	Available on NXTSD104 display ONLY	PR10	Common
			PR11	Normally Open
			PR12	Normally Closed

D) For NXTSD104, if set as General Purpose relay, connections are as follows:

NOTE: If any relay has been previously selected for alarm purpose, power MUST be recycled to reset the relay AFTER changing that relay to general purpose.

Option Parameter	Relay Output	GENERAL PURPOSE Connection Detail for NXTSD104 Only		
17.1	1	Display, Low or Line voltage	PR1	Common
			PR2	Normally Closed
			PR3	Normally Open
17.2	2	Display, Low or Line voltage	PR4	Common
			PR5	Normally Closed
			PR6	Normally Open
17.3	3	Display, Low or Line voltage	PR7	Common
			PR8	Normally Closed
			PR9	Normally Open
17.4	4	PPC6000, PE1 PE4 Line voltage output ONLY		
17.5	5	N/A		
17.6	6	N/A		
17.7	7	NXDBVSD, PZ15, PZ16, Low voltage <50V AC/DC, 200mA		
17.8	8	NXDBVSD, PZ17, PZ18, Low voltage <50V AC/DC, 200mA		
17.9	9	Available on NXTSD104 display ONLY	PR10	Common
			PR11	Normally Closed
			PR12	Normally Open

Option 18.1 to 18.4 – Low Voltage Fail-safe Alarm / Lockout Inputs (0 – 460) LV3

These functions are all fail-safe and may be used to lockout or shutdown the burner.
The fault number generated always relates directly to the input that caused the fault (by going open circuit).
Thus F1 references input 1, F4 references input 4.

The lockout functions themselves are activated by putting a number into option parameters 18.1 to 18.4 for inputs 1 to 4. The number is a one, two or three digit number defined as follows (zero gives no function).



NOTE: In order to enable any Digital Input, a value greater than 0 **MUST** be entered in one of the three columns.

HUNDREDS (fault type)	TENS (fuel type)	UNITS (burner status type)
0 – This alarm will lockout the burner and shows on the display as 'FXX'. Manual intervention (fault mute) is required to restart the burner once the fault condition has cleared. The burner will lockout within one second.	0 or 1 – Fault will be active for any fuel selection (including no fuel selected at all).	0 or 1 – Fault will be active regardless of the current burner status (i.e. all the time).
1 – This alarm will shutdown the burner and show on the display as 'FXX'. When the fault condition clears, this fault will change to 'CXX' and the burner will restart without manual intervention. The burner will shutdown within one second.	2 – Fault will be active only when the currently selected profile fires GAS.	2 – Fault will be active after the fan has started, the air pressure switch has made and the drives have made their purge positions (i.e. once pre-purge starts). Not active during post purge.
2 – This alarm is for indication only. It will appear on the display as 'LXX' but will not stop the burner operating. The limit will operate within three seconds.	3 – Fault will be active only when the currently selected profile fires OIL.	3 – Fault will be active after pre-purge has finished (i.e. drives moving to the ignition position). Not active during post purge.
3 – This operates the same as selection 1 in this column, but the fault will be generated when the input closes. This should not be used for any safety function because if the wire breaks, it cannot be detected.	4 – Fault will be active only when there is a fuel profile currently selected (either firing GAS or OIL).	4 – Fault will be active after pilot ignition has started (status 11 onwards). Not active during post purge.
4 – This alarm will shutdown the burner and show on the display as 'FXX'. When the fault condition clears, this fault will change to 'CXX' and the burner will restart without manual intervention. The burner will shutdown within three seconds.	All other values will work like selection 1 (including zero). 5 to 9 are spare so new functions can be added later.	5 – Fault will be active after main ignition has started (status 13 onwards). Not active during post purge.
5 to 9 are spare and can't be selected.		6 – Fault will be active after the drives have reached their low fire positions and the burner is ready to modulate (status 16). Not active during post purge.
		7 -- fault activates at ignition only (status 10 to 14 inclusive)

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HUNDREDS (fault type)	TENS (fuel type)	UNITS (burner status type)
		8 - fault activates at pre-purge only (status 8)
		All other values will work like selection 1 (including zero). 7 to 9 are spare so new functions can be added later.

NOTE: In order to enable any Digital Input, a value greater than 0 **MUST** be entered in one of the three columns.

Option parameter number	Digital Input Number / Terminals	Fault number
18.1	Input 1 – PA5 to PA11 LOW VOLTAGE supplied from PA11 ONLY	F1 or L1
18.2	Input 2 – PA6 to PA11 LOW VOLTAGE supplied from PA11 ONLY	F2 or L2
18.3	Input 3 – PA7 to PA11 LOW VOLTAGE supplied from PA11 ONLY	F3 or L3
18.4	Input 4 – PA8 to PA11 LOW VOLTAGE supplied from PA11 ONLY	F4 or L4

Option 19.1 to 19.4 – Fail-safe Alarm / Lockout Display Messages LV3

The lockout functions are completely separate from the lockout messages. The messages are chosen from a list of 32 possible items using option parameters 19.1 to 19.4 (for inputs 1 to 4) and option parameters 16.5 & 16.6 for the high voltage inputs (12 & 13). These are the default messages, but at least some of them can be modified using serial communications and a PC.

Option parameter 19.X (or 16.5 / 16.6) value	Meaning
0	Alarm Input Open
1	Low Water
2	Extra Low Water
3	High Water
4	High Gas Pressure
5	Low Gas Pressure
6	High Oil Pressure
7	Low Oil Pressure
8	High Oil Temp.
9	Low Oil Temp.
10	Low Atom. Media
11	High Temperature
12	Low Temperature
13	High Pressure
14	Low Pressure
15	Panel Switch Open
16	High Stack Temp.
17	Fan Interlock
18	End Switch Open
19	Oil Interlock
20	Gas Interlock
21	Forced Lockout
22	High High Water
23	OilGun Interlock

Option parameter 19.X (or 16.5 / 16.6) value	Meaning
24	Low Draft
25	BurnerDoorSwitch
26	Emergency Stop
27	ExitDamperClosed
28 to 31	(spare)



CAUTION

- The product allows for customization of various non-safety critical functions including the modulation control. The option parameters shown below relate to the default modulation control function programmed in the control at the factory. To verify this has not been replaced by an application specific function check with the equipment supplier and / or check option parameter 00.4, if available.

Option 20.0 – Set-point Select input (0 - 35) LV3

This option parameter allows for a digital input to be used to select between modulation setpoint sets 1 and 2 (see option parameters 21.0 and 22.0).

- (1 – 4) Open = Setpoint 1, Closed = Setpoint 2
- (16 or 20) Permanently Closed = Setpoint 2
- (21 – 24) Closed = Setpoint 1, Open = Setpoint 2

NOTE: Options 20.0 to 20.6 and 20.8 have two operational modes. In one case a “high,” level or voltage present, as would be the case of a closed switch, will result in the action specified. In the other case, a “low” level or open switch would cause the action specified.

For example:

- If option 20.1 [boiler shut down] is wired to input 2 (low voltage PA6-PA11) and programmed as 20.1 = 2, the boiler would stay off as long as the switch is closed. In this case if the wire fell off, the boiler would START.
- To reverse this function, option 20.1 would be set as 20.1 = 22. This would require the switch be closed to start the boiler. In this case, if the wire fell off, the boiler would SHUT DOWN.

The table below indicates the value for each of these inputs to affect the desired condition.

Please consider the “broken wire” condition when programming these options.

Option Table 20.0 A

Digital Input Numbers Input CLOSED Powered ON	Digital input used:	Digital Input Numbers Input OPEN Powered OFF
0	None. Input disabled.	20
1	Input 1, PA5 to PA11 LOW VOLTAGE supplied from PA11 ONLY	21
2	Input 2, PA6 to PA11 LOW VOLTAGE supplied from PA11 ONLY	22
3	Input 3, PA7 to PA11 LOW VOLTAGE supplied from PA11 ONLY	23
4	Input 4, PA8 to PA11 LOW VOLTAGE supplied from PA11 ONLY	24

Option 20.1 – Boiler Shutdown input (0 - 35). LV3

- (1 – 4) A high level on the input specified here will cause the boiler to go off and stay off until it is removed. No fault / alarm is generated.
- (16 or 20) **Will cause the burner to SHUT OFF permanently. Should NOT be used.**
- (21 – 24) A low level on the input specified here will cause the boiler to go off and stay off until it is switched high (ON). No fault / alarm is generated.

The input numbers are as specified in Option Table 20.0 A above. This is useful when integration to a building management system is required.

Option 20.2 – Low Fire Hold input (0 - 35). LV3

- (1 – 4) A high level on the input specified here will cause the boiler to go to low fire and stay there until it is removed.
- (16 or 20) **Should NOT be used.**
- (21 – 24) A low level on the input specified here will cause the boiler to go to low fire and stay there until it is switched high (ON).

The input numbers are as specified in Option Table 20.0 A above.

Option 20.3 – Oxygen Trim Disable input (0 - 35). LV3

- (1 – 4) A high level on the input specified here will cause the oxygen trim function to switch off (if it is selected) until it is removed.
- (16 or 20) **Should NOT be used.**
- (21 – 24) A low level on the input specified here will cause the oxygen trim function to switch off (if it is selected) until it is switched high (ON).

The input numbers are as specified in Option Table 20.0 A above. See option parameter 30.5.



Option 20.4 – Ignition Wait input (0 - 35). LV3

- (1 – 4) A high level on the input specified here will prevent the burner from igniting. The drives stay at their ignition positions, but the light up sequence will not start until this input is removed.
- (16 or 20) **Should NOT be used.**
- (21 – 24) A low level on the input specified here will prevent the burner from igniting. The drives stay at their ignition positions, but the light up sequence will not start until this input is switched high (ON).

The input numbers are as specified in Option Table 20.0 A above.

Option 20.5 – Purge Hold input (0 - 35). LV3

- (1 – 4) A high level on the input specified here will prevent the burner moving on past pre-purge. The drives stay at their purge positions, until this input is removed.
- (16 or 20) **Should NOT be used.**
- (21 – 24) A low level on the input specified here will prevent the burner moving on past pre-purge. The drives stay at their purge positions, until this input is switched high (ON).

The input numbers are as specified in Option Table 20.0 A above.

Option 20.6 – Purge Time Start input (0 - 35). LV3

- (1 – 4) A high level on the input specified here will prevent the pre-purge time from starting. The drives stay at their purge positions, until this input is removed and the specified pre-purge time has elapsed.
- (16 or 20) **Should NOT be used.**
- (21 – 24) A low level on the input specified here will prevent the pre-purge time from starting. The drives stay at their purge positions, until this input is switched high (ON) and the specified pre-purge time has elapsed.

The input numbers are as specified in Option Table 20.0 A above.

Option 20.7 – Analog Input 5 function (0 - 2). LV3

This parameter allows analog input 5 (the second modulation input PA12 and PA13 to perform specific functions.

Option parameter 20.7 Value	Digital input used:
0	This input does nothing.
1	Remote tracking input. If the control is in AUTO modulation mode accepting remote inputs, and a valid 4-20mA signal is applied, the control will modulate so as to track this input. 4mA = Low fire, 20mA = High fire. If the signal goes below 3.5mA or above 21mA, this mode will be cancelled and normal PID modulation will resume.
2	Remote Setpoint 1 Input. If the control is in AUTO modulation mode accepting remote inputs, the value of setpoint 1 can be overridden when a valid 4-20mA signal is applied to this input. The zero and span are set in option parameters 21.8 and 21.9. If deviation limits are set (see option parameter 21.5), then the high and low control limits will also move with the setpoint. If the signal goes below 3.5mA or above 21mA, this mode will be cancelled and normal setpoint 1 value will be applied.



Option 20.8 – Digital Input - Lead Boiler Select (0 – 35) LV3

- (1 – 4) A high level on the input specified here will select the lead boiler when closed.
(16 or 20) **Should NOT be used.**
(21– 24) A low level on the input specified here will select the lead boiler when open.

The input numbers are as specified in Option Table 20.0 A above.

**NOTE: For the following Options 21.x and 22.x,
see Section 9.7.10 for PID Tutorial.**

Option 21.0 – Set-point 1 enable (0 - 1) LV1

This option parameter is used to select setpoint 1.

Option 21.1 – Set-point 1 control value (0 - 999 / 00.0 - 99.9 / 0.00 - 9.99) LV1

This is the desired pressure or temperature control value used in the PID control loop for setpoint 1. When the control is in auto mode, the PID control loop will modulate the boiler to maintain the measured value at the same level as set by this parameter.

Option 21.2 – Set-point 1 proportional band (0 - 999 / 00.0 - 99.9 / 0.00 - 9.99) LV1

This is the width of the proportional (modulation) band that is used by the PID control loop for setpoint 1. For example, if the setpoint was 100 psi, and this term was set to 10 psi, then the proportional band would be from 90 psi to 100 psi. A measured value of 90 psi would give high fire, and 100 psi would give low fire, assuming no integral or derivative terms were entered.

A value of 0 means no proportional band – the burner would stay at low fire until the high limit (Cut Out) is reached, then turn off. Proportional control is therefore needed to allow the burner to modulate.

Option 21.3 – Set-point 1 integral time (0 – 999 seconds) LV1

This is the integral time used in the PID control loop for setpoint 1. It may be set to any value from 0 to 999, seconds. If a value of 0 is entered the integral function is disabled, otherwise the time entered is the number of seconds the control will take to give an additional modulation change equal to that currently given by the proportional term. The lower the number (apart from zero), the more affect the integral function has. A large number will cause the integral term to act very slowly.

Integral control is required for the burner to accurately reach its setpoint.

Option 21.4 – Set-point 1 derivative time (0 – 999 seconds) LV1

This is the derivative time used in the PID control loop for setpoint 1. It may be set to any value from 0 to 999. A value of zero will disable the derivative function. A non-zero value will have the effect of ‘advancing’ the modulation rate change caused by a constantly changing measured value by the number of seconds given. A low value will have little effect; a large value will cause a large effect.

Derivative control is seldom needed for boilers, but can improve the response of the modulation system to sudden load changes. Too much can cause control instability.

Option 21.5 - Setpoint 1 control limit type (0 - 2) LV3

This option parameter defines the control limit type for setpoint 1. The control limits are used to automatically turn the burner off when it is not needed, and bring it back on when it is needed.

Option parameter 21.5 value	Meaning
0	No limits. The burner will run until another method is used to switch it off.
1	Absolute limit. The values entered in option parameters 21.6 and 21.7 are the actual limit values.
2	Deviation limit. The values entered in option parameters 21.6 and 21.7 represent a deviation (i.e. offset) from the setpoint 1 control value. This means that if the setpoint control value is changed, the limits are automatically changed correspondingly.

Option 21.6 - Setpoint 1 low limit control value (Cut In) (0 - 999 / 00.0 - 99.9 / 0.00 - 9.99) LV1

If the boiler is off due to a controlled shutdown, this parameter defines the measured value at which the boiler will be turned on again.

Option 21.7 - Setpoint 1 high limit control value (Cut Out) (0 - 999 / 00.0 - 99.9 / 0.00 - 9.99) LV1

If the boiler is on and firing, this parameter defines the measured value at which the boiler will be turned off via a controlled shutdown.

Option 21.8 – Remote Setpoint 1 zero (4ma) value (0 - 999 / 00.0 - 99.9 / 0.00 - 9.99) LV1

This is the zero value for the remote setpoint function specified by option parameter 20.7.

Option 21.9 – Remote Setpoint 1 span (20mA) value (0 - 999 / 00.0 - 99.9 / 0.00 - 9.99) LV1

This is the span value for the remote setpoint function specified by option parameter 20.7.

Option 22.0 – PID (set-point) 2 enable (0 - 1) LV1

This option parameter is used to select setpoint 2.



Option 22.1 – Set-point 2 control value (0 - 999 / 00.0 - 99.9 / 0.00 - 9.99) LV1

This is the control value used in the PID control loop for setpoint 2. When the control is in auto mode, the PID control loop will modulate the boiler to maintain the measured value at the same level as set by this parameter.

Option 22.2 – Set-point 2 proportional band (0 - 999 / 00.0 - 99.9 / 0.00 - 9.99) LV1

This is the width of the proportional (modulation) band that is used by the PID control loop for setpoint 2. For example, if the setpoint was 100 psi, and this term was set to 10 psi, then the proportional band would be from 90 to 100 psi. A measured value of 90 psi would give high fire, and 100 psi would give low fire, assuming no integral or derivative terms were entered.

A value of 0 means no proportional band – the burner would stay at low fire until the high limit (Cut Out) is reached, then turn off. Proportional control is therefore needed to allow the burner to modulate.

Option 22.3 – Set-point 2 integral term (0 – 999 seconds) LV1

This is the integral time used in the PID control loop for setpoint 2. It may be set to any value from 0 to 999, seconds. If a value of 0 is entered the integral function is disabled, otherwise the time entered is the number of seconds the control will take to give an additional modulation change equal to that currently given by the proportional term. The lower the number (apart from zero), the more affect the integral function has. A large number will cause the integral term to act very slowly.

Integral control is required for the burner to accurately reach its setpoint.

Option 22.4 – Set-point 2 derivative term (0 - 100) LV1

This is the derivative time used in the PID control loop for setpoint 2. It may be set to any value from 0 to 999. A value of zero will disable the derivative function. A non-zero value will have the effect of 'advancing' the modulation rate change caused by a constantly changing measured value by the number of seconds given. A low value will have little effect; a large value will cause a large effect.

Derivative control is seldom needed for boilers, but can improve the response of the modulation system to sudden load changes. Too much can cause control instability.

Option 22.5 - Setpoint 2 control limit type (0 - 2) LV3

This option parameter defines the control limit type for setpoint 2. The control limits are used to automatically turn the burner off when it is not needed, and bring it back on when it is needed.

Option parameter 22.5 value	Meaning
0	No limits. The burner will run until another method is used to switch it off.
1	Absolute limit. The values entered in option parameters 22.6 and 22.7 are the actual limit values.
2	Deviation limit. The values entered in option parameters 22.6 and 22.7 represent a deviation (i.e. offset) from the setpoint 1 control value. This means that if the setpoint control value is changed, the limits are automatically changed correspondingly.



Option 22.6 - Setpoint 2 low limit control value (0 - 999 / 00.0 - 99.9 / 0.00 - 9.99) LV1

If the boiler is off due to a controlled shutdown, this parameter defines the measured value at which the boiler will be turned on again.

Option 22.7 - Setpoint 2 high limit control value (0 - 999 / 00.0 - 99.9 / 0.00 - 9.99) LV1

If the boiler is on and firing, this parameter defines the measured value at which the boiler will be turned off via a controlled shutdown.

Option 22.8 – Maximum modulation rate

DEFAULT 0 – unlimited / Range 1 – 100%

Values represent: 0 = unlimited
 1 – 100 is maximum %

Option 22.9 – *Reserved*

Option 23.0 – Warming Enable (0 or 1) LV1

This parameter allows a warming function to be applied to the boiler, and makes option parameters 23.1 and 23.2 available. If zero is entered, no warming limit is applied.

Option 23.1 – Warming Limit (0 - 999 / 00.0 - 99.9 / 0.00 - 9.99) LV1

If, when the burner starts up and reaches modulation, the measured pressure / temperature value is lower than that specified here, the control will hold the boiler at low fire until the value specified is reached. The control will hold the boiler at low fire (in status 15) indefinitely unless a non-zero time has been entered in option parameter 23.2.

Note: If during normal modulation, the measured value falls below this limit having once been above it, the warming function will not be re-applied. The warming function is only applied on a burner start-up.

Option 23.2 – Warming Time (0 to 999 minutes) LV1

This parameter specifies a maximum time to hold the burner at low fire for, before ignoring the warming limit and allowing normal modulation.

Additionally, when the burner goes off (for any reason other than power interruption), and the measured value falls below the warming limit, the warming function will not be activated again until this time has passed.

NOTE: Further parameters (up to 29.9) may be available here if they are part of a users program or non-standard manufacturers program.